

Sketch: Application and Adaptive Optimization of Continuous-Time Liquid Neural Networks in Language Modeling

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1 Research Motivation

Traditional Long Short-Term Memory (LSTM) networks are discrete-time series models whose states are updated with fixed integer steps, making them susceptible to gradient fluctuations and less capable of modeling long-term dependencies in long texts. Liquid Neural Networks (Liquid NNs), on the other hand, are continuous-time dynamic systems whose states evolve smoothly over time, better aligning with the asymptotic nature of language context. A fundamental difference lies in their temporal modeling paradigms: traditional static positional encoding is designed for discrete time, leading to a mismatch between the temporal representation and the continuous-time representation of Liquid NNs, resulting in structural mismatch and performance limitations. This project validates the effectiveness of the continuous-time Liquid NN paradigm in language modeling and designs a lightweight adaptive optimization scheme.

2 Research Methodology

A strict controlled variable design of "baseline $\rightarrow$ validation $\rightarrow$ optimization" is adopted. All models share exactly the same basic architecture (embedding dimension=128, hidden dimension=256, vocabulary size 5k), training objective (next-token prediction) and hyperparameters, with only two independent variables modified to ensure interference-free comparison:

- Baseline Model: LSTM + standard static positional encoding;
- Validation Model: LiquidLayer + standard static positional encoding, used to verify the inherent modeling capability of the continuous-time paradigm;
- Optimization Model: LiquidLayer + dynamic time-aware positional encoding, which resolves the temporal misalignment between the continuous-time paradigm and discrete encodings.

3 Experimental Methodology

3.1 Dataset

The standard language modeling dataset WikiText-2 (approximately 2 million tokens) is adopted. With a larger scale than course experiment data and longer contextual texts, it ensures academic comparability. The dataset is split into training and validation sets according to the official partition, and further divided into short sequences (20 tokens) and medium-length sequences (21–50 tokens) by length, which are specifically used to evaluate long-distance dependency modeling capability.

3.2 Training Setup

The AdamW optimizer and cross-entropy loss are uniformly used, with fixed batch size and learning rate, and early stopping is adopted to avoid overfitting. Each model is independently trained 3 times repeatedly, and the

average value is taken to reduce the impact of random initialization, ensuring stable and reliable results.

3.3 Evaluation Framework

The core metrics recognized in the field of language modeling are adopted for comprehensive evaluation from three dimensions: capability, stability and efficiency:

- **Perplexity (PPL):** The most core metric, which directly measures the model's ability to predict linguistic sequences, with lower values indicating higher modeling accuracy;
- **Training Loss Volatility:** Reflects the stability of the model during the training process, demonstrating the advantage of continuous-time state evolution;
- **Convergence Epochs:** Measures the training efficiency required for the model to reach stable performance, reflecting the learning characteristics of different temporal modeling paradigms.

At the same time, the performance differences of models on short/medium sequences are compared, focusing on verifying the potential advantages of the continuous-time paradigm in long-term dependency modeling.

4 Research Value

This project is not a simple replacement of network structures, but a fundamental comparison of the differences between discrete and continuous-time modeling paradigms for natural language. The credibility of the conclusions is guaranteed through standardized datasets and multi-dimensional core metrics.